

Experimental Data:

Table 1: For the determination of the thickness of the lens.

No. of Obs.	Main scale reading, M (cm)	Vernier scale reading, V	Vernier constant, C (cm)	Value of Vernier reading, $F = V \times C$ (cm)	Total reading, $(M + F)$	Average reading, t (cm)	Reading of the mid-point of the lens, $\frac{t}{2}$ (cm)
1.	0.5	4	0.01				
2.	0.5	5					
3.	0.5	6					

Table 2: For the determination of height (h).

No. of Obs.	Which surface	Linear scale reading, M (cm)	Circular scale reading, V	Least count, C (cm)	Value of the circular reading, $F = V \times C$ (cm)	Total reading, ($M + F$)	Average reading, x (cm)	Height, $h = x_2 - x_1$ (cm)
1.	On the plane mirror (x_1)	0	19	0.001			$x_1 =$	
2.		0	20	0.001				
3.		0	21	0.001				
1.	On the lens (x_2)	0.1	83	0.001			$x_2 =$	
2.		0.1	84	0.001				
3.		0.1	85	0.001				

Table 3: For the determination of the values f_1 and F .

No. of Obs.	Height, h_1 (cm)	Average height, h_1 (cm)	Focal length, $f_1 = h_1 + \frac{t}{2}$ (cm)	Height, h_2 (cm)	Average height, h_2 (cm)	Focal length, $F = h_2 + \frac{t}{2}$ (cm)
1.	21.1			34.2		
2.	21.1			34.1		
3.	21.3			34.3		

Determination of the distance, D between the legs of the spherometer:

$$D = \frac{D_1 + D_2 + D_3}{3} = \frac{4.3 + 4.3 + 4.3}{3} = 4.3 \text{ cm}$$

Calculations:

- $r_1 = \left(\frac{D^2}{6h} + \frac{h}{2} \right) = \dots\dots\dots \text{cm}$
- $f_2 = \frac{F f_1}{F - f_1} = \dots\dots\dots \text{cm}$
- $\mu = 1 + \frac{r_1}{f_2} = \dots\dots\dots$

Results:

Refractive index of experimental liquid =